

Desynchronization of Circadian Rhythms in Winterover Antarctic Expeditioners

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Abstract

Four male subjects participated in a field study at Maitri station for 14 months, for the investigation of the circadian rhythms of oral temperature (OT), heart rate (HR), blood pressure (BP) and hand grip strength (GS). Each subject was studied in five sessions at the Indian base station at Antarctica and once during return journey in the ship. Four hourly records of the above said parameters were obtained for 24 hr. The psychoemotional status was observed by standard questionnaire. The study revealed apparent circadian rhythm alterations in OT, HR, BP and GS during winters. In particular the OT and HR rhythms were adjusted by the end winters where as the others failed to return to normal rhythmic state even in summer following at Antarctica. The psychoemotional status correlated with altered circadian rhythms of the body.

Introduction

Under natural photoperiodic conditions of 12:12 light dark cycle in human being there occurs well established circadian rhythms of various physiological, immunological and psychological parameters (Comelissen *et al.*, 1989, Pati *et al.*, 1987, Reinberg *et al.*, 1988, Gupta & Pati, 1994). Most of these studies performed in human circadian rhythms have been restricted to temperate regions of the world. The different climatic conditions have been reported to modulate the process of rhythm synchronization and desynchronization. Reinberg *et al.* (1984) have shown that subjects sojourning in high arctic (79°N) during summer show internal desynchronization among circadian rhythms. Such internal desynchronization has also been reported in isolation experiments (Aschoff *et al.*, 1982) and shift workers (Reinberg *et al.*, 1988, 89). However, it is not well documented how much the Antarctic conditions of extended light

dark cycles with isolation could influence the circadian rhythms of human subjects. The present study was thus conducted on members of 12th Indian Antarctic expedition during their winter stay of 14 months at Maitri station to determine the effect of altered circadian rhythms, if any, on psychological or behavioural changes. The self assessing questionnaires were used to determine the anxiety and depression status of the subjects.

Materials and Methods

The study was conducted on 4 healthy male winterover Antarctic Expedition members aged 39.5 ± 1.5 yr, between January 1992 and March 1994. All the four subjects were from army. The subjects volunteered for the study willingly and they were instructed to abstain from liquor on the day of study. On 6 occasions following physiological variables, *e.g.*, oral temperature (OT), arterial blood pressure (BP), heart rate (HR) and right hand grip strength (GS) were recorded every 4 hourly for 24 hr. Five recording sessions, at 3 monthly intervals were conducted at Indian base camp Maitri and one during return journey on the ship.

The OT was measured using a standard clinical thermometer (celsius), HR from radial pulse counting for 1 min. using a stop watch, sitting BP from mercury sphygmo-manometer. The GS was measured using hand grip dynamometer, in a standardized body and arm position, sitting with arm and forearm hanging vertically along the side of body. For each GS record 3 measurements were made and the highest of the 3 figures was considered for analysis. The actual time in hr and min was recorded for measurement of all variables which were performed 7 times per 24 hr. The selection of such sampling scheme was based both on feasibility within field conditions of Antarctic station and on results from comparative analysis of other workers report on human circadian rhythms (Motahasi *et al.*, 1987, Pati & Saini, 1991).

In addition to this psychological studies were performed on these members using questionnaires to assess anxiety (STAI Form Y-I developed by Charles D. Spielberg California) and depression state. During data collection period the subjects had no illness and were taking no medication. The subjects were allowed to sleep *ad libitum*.

Statistical Analysis

For this study cross correlations and ANOVA has been used for analysis of periodicity significance. This method has its own limitations therefore cosinar method has also been used to provide complementary results.

Results

Heart rate circadian rhythm: The pooled data of circadian rhythm of 4 subjects is depicted graphically in Fig.1. In our study the circadian rhythm of HR was not significant during summer (Jan and March) and peak winter (June). However, the HR rhythm was prominent and significant in end winters (Sep. $p < 0.01$ and Dec. $p < 0.001$). It was also noticed that acrophase location which occurred in mid afternoon (14 hr) did not change throughout the period of study. Thus indicating no acrophase shift.

Oral temperature rhythm: The pooled average of OT of all the subjects showed statistically significant rhythm only during early summers, *i.e.*, Jan. and Dec. Non-circadian rhythms was detected in end summers and winters. However, when acrophases were compared by cosinar analysis at different seasons of Antarctica, it revealed that except during end summers, the peak in oral temperature occurred at about 1800 hr. In comparison during end summer the OT peak was located at night *i.e.* between 2200 & 0200 hr (Fig.2).

Blood pressure rhythm: On statistical rhythm analysis the cosinar curves as well as ANOVA revealed that except during return journey in March the systolic BP and in early summer (Jan) the diastolic BP rhythms were significant ($p < 0.01$). Whereas during remaining period of summer and winters no rhythmicity was recorded in systolic and diastolic BP (Figs. 3 & 4). The prominent feature of systolic BP was a peak rise in the first phase of night (2200 hr). However, no such consistent acrophase was recorded for diastolic BP (Fig. 4).

Grip strength rhythm: Non-circadian rhythm of hand grip strength was detected throughout the period of summer and winter stay. Furthermore, it was also revealed that there was no fixed acrophase location for grip strength (Fig. 5).

Psychoemotional status: Results as shown in Table 1, anxiety state, trait and depression scores of the members in winters were higher, though the difference in comparison to summer was not significant. The results showed a change in score related to period of stay and photoperiod. It was found that low anxiety score was most pronounced in Dec., *i.e.*, mid summer.

Discussion

Results of this study clearly demonstrate that physiological variables under investigation lost their circadian rhythmicity in subjects during Antarctic winter stay. Our findings are consistent with earlier reports of alteration of circadian rhythmicity of human subjects during their stay in Antarctic winters (Kennaway

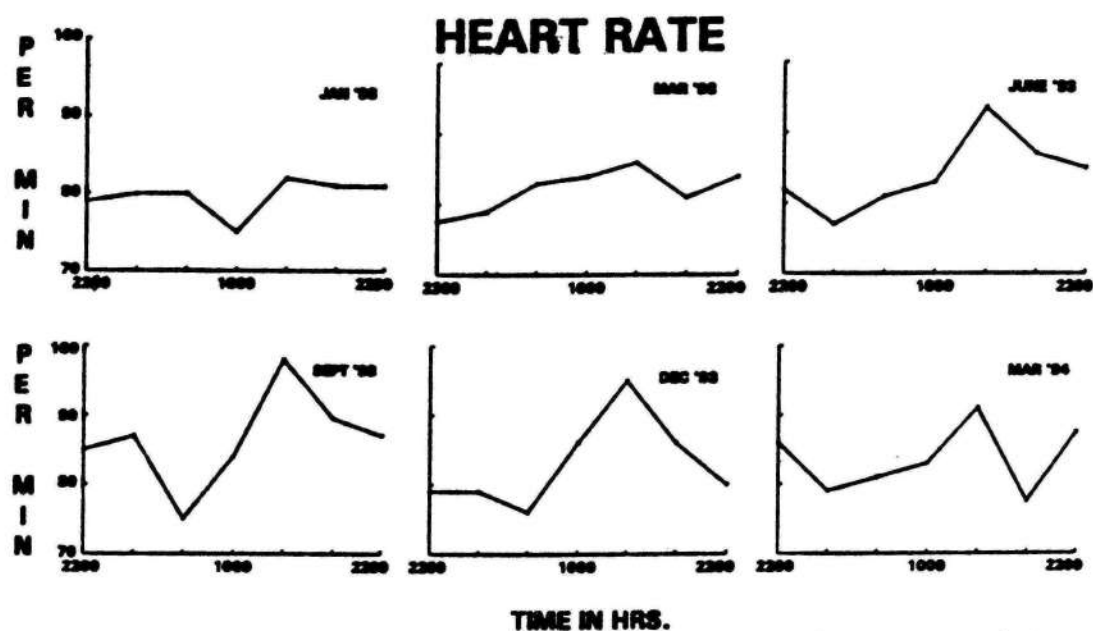


Fig 1: Circadian rhythm of heart rate was non-significant throughout except end winter (Sept.) and start summer (Dec.). The acrophase was consistent at 1400 hr during summer and winter

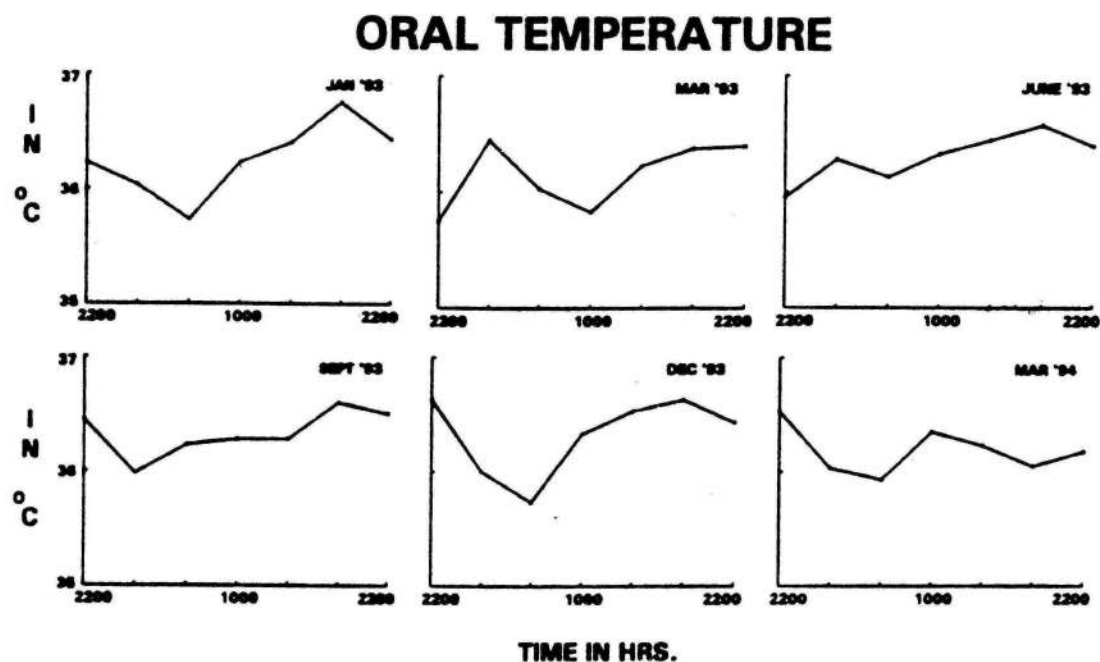


Fig 2: Circadian rhythm of oral temperature was significant ($p < 0.001$) only in summer. There was phase delay (acrophase at 0200 hrs instead of 1800 hrs) during end summer and winter

et al., 1991). The heart rate and oral temperature rhythms were the only one which exhibited significant rhythmicity either in late winters and summer respectively. The data also shows that during peak winters even these robust rhythms were lost. In fact what is most interesting is that although the rhythmicity was lost for HR, the pattern of cycle even during winters remained

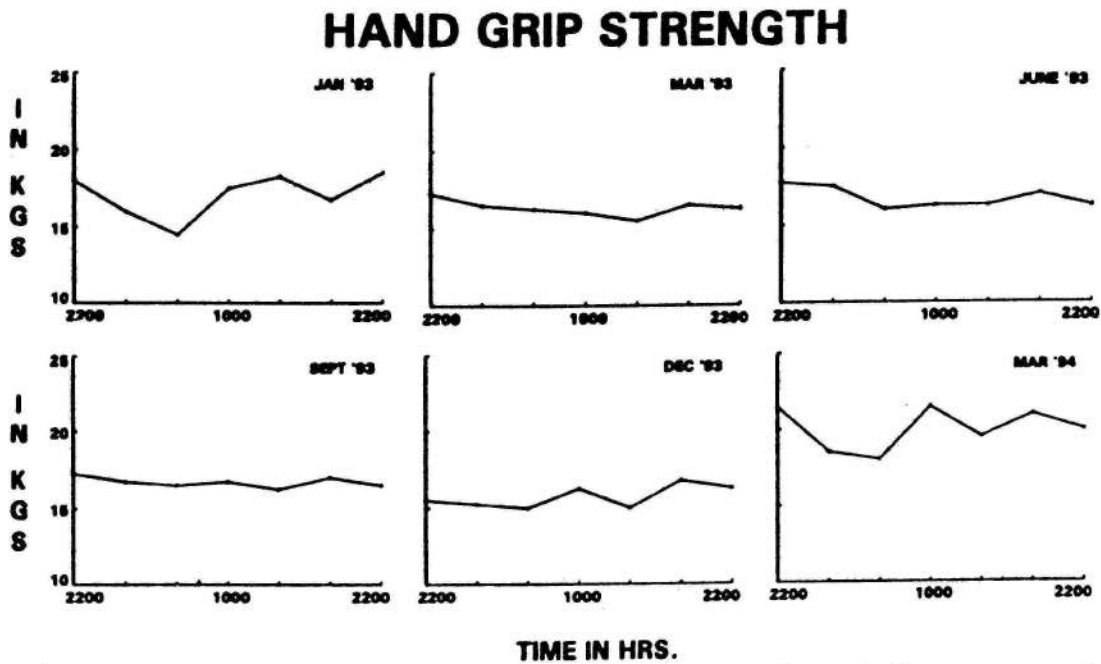


Fig 5: Circadian rhythm of hand grip strength was non-significant during summer and winter. The acrophase was also not consistent throughout

Table 1: Mean and SD of Anxiety and Depression Scores of Wintering Subjects during Antarctic Summer and Winter

Sessions	Anxiety state	Anxiety trait	Depression
Basal	24.00±3.79	26.20±3.49	25.40±4.22
End Summer	25.71±5.41	26.00±4.34	23.85±3.44
Mid WinterR	27.71±7.20	28.42±6.77	No Recording
End Winter	29.00±8.81	28.28±7.02	26.43±5.20
Early Summer	22.17±3.18	23.66±3.29	No Recording
Return	23.85±5.66	26.28±6.27	23.57±3.62

and following December, the members were required to conduct outdoor activity in the camp. Generally, the summer team departs in March and the wintering subjects perform the work of looking after the stores as well as completing the outdoor construction work before the winter starts. Similarly during December month, as the expedition is towards the end, there is increased outdoor activity for handing over of the station to the new members. The possibility is that increased outdoor physical activity promoted the phase shift of the oral temperature rhythm. The internal desynchronization of temperature and several other circadian rhythms have been demonstrated which are related to physical activity status of the individual (Bicakova *et al.*, 1981, Bicakova

et al., 1984). The present study did not involve strict adherence to sleep-awake schedule, which is an important contributing factor for altering several physiological rhythms (Moore Ede MC *et al.*, 1980).

Alterations and disturbances of sleep were reported in these subjects. We have collected the sleep data which is also being sent as brief communication in this report.

In earlier studies by other workers, it has been demonstrated that under constant conditions the body temperature rhythm maintains a fixed phase relationship (Shanahan and Czeisler, 1991). Our results support the conclusion that in humans under non-constant field conditions the endogenous temperature rhythm is superimposed and masked by factors such as light-dark cycle, exercise, metabolic, sleep-awake schedules.

As regards the rhythmicity of other physiological variables studied herein we recorded non-significant rhythms throughout the period of Antarctic stay. It appears that photoperiodism, day length and different levels of illumination also play an important role in rhythm impairment. There was substantial phase shift and pattern change in diastolic blood pressure and hand grip strength rhythm. However, the systolic blood pressure was non-rhythmic, but not associated with phase shifting as was also recorded for HR. This fundamental relationship between the systolic blood pressure and HR was striking in our data. The alterations in systolic BP is the direct reflection of cardiac activity is apparent in our data. The diastolic blood pressure on the other hand, indicates the magnitude of vascular sympathetic tone and showed a discrete response.

We have further demonstrated that anxiety and depression state of winter-over personnel were directly related to circadian rhythm changes. In other words, during December *i.e.*, early summer months most of circadian rhythms had returned to near normal and the anxiety state and trait also reduced. The finding of this study still raises the obvious question that lowered anxiety state could have been due to job satisfaction, presence of summer team and near completion of expedition. The answer to this question whether the psychoemotional status was a casual or noncasual response of impaired biological clock cannot be unequivocally answered by this study. It should be here emphasized that in this study the sleep-awake cycle nor the physical activity status were controlled. Thus this obviously was not entirely satisfactory controlled investigation. In future we shall be more particular to control the exercise, meal and sleep-awake schedules which seem to be important synchronizing factors for circadian rhythms.

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